

Topics to discuss

- Composite Trapezoidal Rule
- Composite Simpson's $\frac{1}{3}$ & $\frac{3}{8}$ Rule
- Error in Integration
- Numerical Problem

Difference between Trapezoidal and Composite Trapezoidal Rule

The difference is mainly how many subintervals (n) we use.

Trapezoidal Rule:

This is the basic formula where the whole interval $[a, b]$ is treated as one single trapezium.

$$\int_a^b f(x) dx = \frac{h}{2} [f(a) + f(b)]$$

where, $h = \frac{b-a}{n}$, $n=1$

Here, h is step length

n is no. of interval

In this formula, only two points are used

$$x_0 = a, \quad x_1 = b$$

Composite Trapezoidal Rule :

In this method, the interval $[a, b]$ is divided into many smaller equal parts, and then the trapezoidal rule is applied to each part. After that all areas are added.

If the interval is divided into n equal subintervals, then

step size, $h = \frac{b-a}{n}$, and formula becomes,

$$\int_a^b f(x) dx = \frac{h}{2} \left[f(x_0) + 2[f(x_1) + f(x_2) + \dots + f(x_{n-1})] + f(x_n) \right]$$

OR

$$= \frac{h}{2} \left[y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n \right]$$

In simple terms, the main difference:

* Trapezoidal Rule:

→ Uses only one trapezoid ($n=1$)

→ less accurate

* Composite Trapezoidal Rule:

→ Uses many trapezoids ($n=n$)

→ More accurate

Difference between Simpson's $\frac{1}{3}$ rd and Composite Simpson's $\frac{1}{3}$ rd Rule

The main difference is number of subinterval used.

In Simpson's $\frac{1}{3}$ rd Rule :

Here the integration is done over one pair of subintervals only.

That means the interval is divided into 2 equal parts.

Formula :
$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [y_0 + 4y_1 + y_2]$$

Here,
$$h = \frac{x_2 - x_0}{n}, \quad n = 2$$

Only 3 ordinates are used: y_0, y_1, y_2

Composite Simpson's $\frac{1}{3}$ rd Rule :

Here full interval is divided into many even subintervals

formula :

$$\int_{x_0}^{x_n} f(x) dx = \frac{h}{3} \left[y_0 + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2}) + y_n \right]$$

Here, $h = \frac{x_n - x_0}{n}$, n must be even

Difference between Simpson's $\frac{3}{8}$ and Composite Simpson's $\frac{3}{8}$ Rule

Simpson's $\frac{3}{8}$ Rule :

Here the integration is done over 3 equal subintervals only

Formula :

$$\int_{x_0}^{x_3} f(x) dx = \frac{3h}{8} [y_0 + 3(y_1 + y_2) + y_3]$$

Here, $n = 3$

4 ordinates are used: y_0, y_1, y_2, y_3

Composite Simpson's $\frac{3}{8}$ Rule :

Here, the interval is divided into many subintervals,

Formula :

$$\int_{x_0}^{x_n} f(x) dx = \frac{3h}{8} \left[y_0 + 3(y_1 + y_2 + y_4 + y_5 + \dots) \right. \\ \left. + 2(y_3 + y_6 + y_9 + \dots) + y_n \right]$$

Here, n must be multiple of 3

Numerical Problem :

Q: Find the approximate value of the integral

$$\int_0^6 \frac{1}{1+x^2} dx$$

using (a) Trapezoidal Rule

(b) Simpson's $\frac{1}{3}$ Rule

(c) Simpson $\frac{3}{8}$ Rule

solution, taking $n=6$

$$\text{step length, } h = \frac{6-0}{6} = 1.$$

lets make table for, $f(x) = \frac{1}{1+x^2}$.

x	0	1	2	3	4	5	6
$f(x)$	1	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{17}$	$\frac{1}{26}$	$\frac{1}{37}$

Trapezoidal Rule:

$$\begin{aligned}\int_{x_0}^{x_6} f(x) dx &= \frac{h}{2} \left[y_0 + 2(y_1 + y_2 + y_3 + y_4 + y_5) + y_6 \right] \\ &= \frac{1}{2} \left[1 + 2\left(\frac{1}{2} + \frac{1}{5} + \frac{1}{10} + \frac{1}{17} + \frac{1}{26}\right) + \frac{1}{37} \right] \\ &= 1.410798 \quad \underline{A_1}\end{aligned}$$

Simpson $\frac{1}{3}$ rd Rule:

$$\begin{aligned}\int_{x_0}^{x_6} f(x) dx &= \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) + y_6] \\ &= \frac{1}{3} \left[1 + 4\left(\frac{1}{2} + \frac{1}{10} + \frac{1}{26}\right) + 2\left(\frac{1}{5} + \frac{1}{17}\right) + \frac{1}{37} \right] \\ &= 1.366173\end{aligned}$$

Simpson $\frac{3}{8}$ th Rule:

$$\begin{aligned}\int_{x_0}^{x_6} f(x) dx &= \frac{3h}{8} [y_0 + 3(y_1 + y_2 + y_4 + y_5) + 2y_3 + y_6] \\ &= \frac{3}{8} \left[1 + 3\left(\frac{1}{2} + \frac{1}{5} + \frac{1}{17} + \frac{1}{26}\right) + 2 \times \frac{1}{10} + \frac{1}{37} \right] \\ &= 1.3570808 \text{ etc.}\end{aligned}$$

Exact Value,

$$\int_0^6 \frac{1}{1+x^2} dx = \left[\tan^{-1} x \right]_0^6 = \tan^{-1}(6) - \tan^{-1}(0) = 1.405647$$

Hence,

$$\text{Exact Value} = 1.405647$$

$$\text{Trapezoidal Rule} = 1.410798$$

$$\text{Simpson } \frac{1}{3}^{\text{rd}} \text{ Rule} = 1.366173$$

$$\text{Simpson } \frac{3}{8}^{\text{th}} \text{ Rule} = 1.3570808$$

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